COURSE TITLE : PAINTS AND SURFACE COATINGS
COURSE CODE : 6094
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 60/6
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	Light and Components	15
2	Classification of Paints	15
3	Methods of painting and adhesion.	20
4	Mechanism of film formation and properties of coatings	10
TOTAL		**60**

COURSE OUTCOME

Sl.	Sub	Student will be able to
1	1	To understand the properties of light. T
	2	To understand the components of paints and its preparations.
2	1	To identify the different types of paints and its classification.
	2	To understand and apply the different methods of Painting and adhesion of coatings.
3	1	To know the mechanism of film formation and the various properties of coatings

SPECIFIC OUTCOME

MODULE I

1.1.0 Light and Components of paints
1.1.1 Explain Reflection, Refraction and Diffraction
1.1.2 Distinguish between additive and subtractive colour mixing
1.1.3 Explain specular gloss and bloom gloss
1.1.4 List out the components of paints
1.1.5 Explain the properties of pigments
1.1.6 Describe the preparation of pigment dispersion
1.1.7 Explain briefly extenders, solvents and dilutents
1.1.8 Describe types of solvents and solvent properties

1.1.9 Explain additives affecting viscosity
1.1.10 Explain the preparation of paint

MODULE II

2.1.0 Classification of paints
2.1.1 Explain polymeric resin based paints
2.1.2 Distinguish between oil paint and emulsion paint
2.1.3 Describe brieflyacrylics paints, epoxy coatings, polyurethanes, silicones, formaldehyde based resins, chlorinated rubbers and hydrocarbon based resin.
2.1.4 Explain the classification of paints briefly based on application.
    2.1.4.1 automotive finishes
    2.1.4.2 Coil coatings
    2.1.4.3 Can coatings
    2.1.4.4 Marine coating
    2.1.4.5 Aircraft finishes

MODULE III

3.1.0 Methods of Surface Cleaning and Adhesion Properties of Coatings
3.1.1 Explain the methods adopted for surface cleaning
3.1.2 Describe the methods of application of paints
    3.1.2.1 Brushing
    3.1.2.2 Dip Coating
    3.1.2.3 Flow Coating
    3.1.2.4 Roller Coating
    3.1.2.5 Spray Painting
    3.1.2.6 Electro deposition
    3.1.2.7 Chemiphoretic Deposition
3.1.3 Explain adhesion Properties of Coatings
3.1.4 Explain the factors Affecting Adhesive Bond

MODULE IV

4.1.0 Surface Preparation and Painting
4.1.1 Explain the mechanism of film formation
4.1.2 Describe physical drying, oxidative drying and chemical drying.
4.1.3 Explain the factors affecting coating properties like.
    4.1.3.1 Film Thickness
    4.1.3.2 Film Density
    4.1.3.3 Internal Stresses

4.1.3.4 Pigment Volume Concentration (PVC).

4.1.4 Explain the different Methods used for Film Preparation

4.1.5 Explain the properties of coatings Barrier Properties, Mechanical Properties Optical Properties, Color, Gloss, Hiding Power and Ageing Properties.

4.1.6 Explain the properties such as Floating, Silking, Cratering, Foaming, Skinning, Flame Retardance, Slip Resistance and Storage Stability

CONTENT DETAILS

MODULE I

Light: Reflection, Refraction, Diffraction, Color Science, Additive Color Mixing, Subtractive Color Mixing, Gloss-

Introduction: Components of Paints- - Pigments -Manufacture - Pigment Properties - - Different Types- Selection, Dispersion and Color Matching of Pigments- Pigment Dispersion and Preparation, Extenders, Solvents, Different Types, Solvent Properties, Oil, Driers, Resins, Dilutents, Additives Affecting: Viscosity Interfacial Tensions- Paint Preparation- Formulation.

MODULE II

Classification Based on Polymeric Resin, Emulsion, Oil and Alkyd Paints, crylic P N Classification Based on Polymeric Resin, Emulsion, Oil and Alkyd Paints, Acrylic Paints, Epoxy Coatings, Polyurethanes, Silicones, Formaldehyde based Resins, Chlorinated Rubbers, Acrylics, and Hydrocarbon resins. Classification based on Application. Fluoro polymers, Vinyl Resins, Appliance Finishes, Automotive Finishes, Coil Coatings, Can Coatings, Marine Coatings, and Aircraft Finishes.

MODULE III

Surface Cleaning Methods, Chemical Conversion Treatments, Paint Application, Brushing, Dip Coating, Flow Coating, Roller Coating, Spray Painting, Electro deposition, Chemiphoretic Deposition- Adhesion Properties of Coatings, Factors Affecting Adhesive Bond, Properties such as Floating, Silking, Cratering, Foaming, Skinning, Flame Retardance, Slip Resistance and Storage Stability. Coatings, Polyurethanes, Silicones, Formaldehyde basResinlorinated

MODULE IV

Mechanism of Film Formation, Physical Drying, Oxidation Drying, Chemical Drying, Factors Affecting Coating Properties, Film Thickness, Film Density, Internal Stresses, Pigment Volume Concentration (PVC). Different Methods used for Film Preparation, Barrier Properties, Mechanical Properties and Optical Properties of Coatings, Color, Gloss, Hiding Power, Ageing Properties, Factors Affecting Viscosity of Paints,.

REFERENCE BOOKS

1. Swaraj Paul, “Surface Coatings: Science and Technology”, Wiley – Interscience 1985.
2. R.Lambourne. “Paint and Surface Coatings – Theory and Practice”, Ellis Horwood Chichester 1987.
3. Turner, Paints and Surface Coating